Jake Brawer, Ph.D.

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RESEARCH OVERVIEW

Summary — I develop **collaborative AI systems** that integrate perception, reasoning, and learning to support seamless human–robot interaction. My research combines advances in machine learning with foundational ideas from artificial intelligence and cognitive science to develop agents that use **structured, abstract knowledge** to make social and physical interactions more **robust, transparent, and natural**.

PROFESSIONAL EXPERIENCE

2023-Current — **Postdoctoral Researcher**
University of Colorado, Boulder, Advisors: Alessandro Roncone and Bradley Hayes
Leading project on human–robot teaming, assistive robotics, and decision support.

Spring 2024 — **Lecturer, Advanced Robotics** (CSCI 4302/5302)
University of Colorado, Boulder
Sole lecturer for graduate-level robotics course covering motion planning, SLAM, optimization, and reinforcement learning.

EDUCATION

2016-2023 — **Ph.D. in Computer Science** *Yale University*, Advisor: Brian Scassellati

2012-2016 — **B.A. in Cognitive Science; Computer Science minor** *Vassar College*

SKILLS

Programming	**Python** , **C-C++**, **C#, LaTeX**, **Git**, **Jekyll**, **Emacs**, **Docker**, **Unity**
ML/AI/CV Tools	**Scikit-learn**, **SciPy**, **Pytorch**, **numpy**, **pandas**, **matplotlib**, **NLTK**, **ROS**, **Gazebo**, **RViz**, **Gurobi**, **Prolog**, **PDDL**, **OpenCV**, **Open3D**
Robots and Sensors	**Baxter**, **Sawyer**, **UR5**, **Kinect**, **RealSense**, **RGB-D** cameras
System	10+ years of daily **Linux** experience
Soft Skills	**Strong communicator**, **Loves collaboration**, **Fast learner**, **Leadership**

JOURNAL ARTICLES

Shivendra Agrawal, Ashutosh Naik, **Jake Brawer**, Alessandro Roncone, and Bradley Hayes (2025). "**ShelfAware: Semantic Particle Filter Localization with Low-Cost Sensors**". In preparation. *A novel semantic Monte Carlo localization algorithm that requires just an RGB-D camera and a visual odometry camera.*

Meiying Qin, **Jake Brawer**, and Brian Scassellati (2022a). "**Robot Tool Use: A Survey**". In: *Frontiers in Robotics and AI* 9, p. 369.

* Meiying Qin, * **Jake Brawer**, and Brian Scassellati (2021). "**Rapidly Learning Generalizable and Robot-Agnostic Tool-Use Skills for a Wide Range of Tasks**". In: *Frontiers in Robotics and AI* 8, p. 380.

Jake Brawer, Aaron Hill, Ken Livingston, Eric Aaron, Joshua Bongard, and John H Long Jr (2017). "**Epigenetic Operators and the Evolution of Physically Embodied Robots**". In: *Frontiers in Robotics and AI* 4, p. 1.

Conference proceedings

Jake Brawer, Clare Lohrmann, Que Liu, Anthony Ries, Alessandro Roncone, and Bradley Hayes (2025). "**Effort-Aware Multi-Agent Task Allocation in Sequential Collaborative Scenarios**". In preparation. *A decision support system can assist human users in regulating physical and cognitive effort during collaborative task sequences with autonomous agents.*

Gyanig Kumar, Yashwant Gandham, Yi-Shiuan Tung, Anthony Ries, **Jake Brawer**, Alessandro Roncone, and Bradley Hayes (2025). "**Flexible Multimodal Shared Control for Assistive Robot Arm Manipulation**". In preparation. *A multimodal shared autonomy system that combines eye gaze, joystick input, and semantic scene understanding to infer user intent and dynamically generate task-aligned control modes for assistive teleoperation.*

Nikhil Hulle, Stéphane Aroca-Ouellette, Anthony J Ries, **Jake Brawer**, Katharina von der Wense, and Alessandro Roncone (2024). "**Eyes on the Game: Deciphering Implicit Human Signals to Infer Human Proficiency, Trust, and Intent**". In: *33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2024)*. Accepted for publication, proceedings pending.

Jake Brawer, Debasmita Ghose, Kate Candon, Meiying Qin, Alessandro Roncone, Marynel Vazquez, and Brian Scassellati (2023). "**Interactive Policy Shaping for Human-Robot Collaboration with Transparent Matrix Overlays**". In: *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*. IEEE.

Meiying Qin, **Jake Brawer**, and Brian Scassellati (2022b). "**Task-Oriented Robot-to-Human Handovers in Collaborative Tool-Use Tasks**". In: *2022 31st IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*. IEEE.

Jake Brawer, Meiying Qin, and Brian Scassellati (2020). "**A causal approach to tool affordance learning**". In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, pp. 8394–8399.

Brian Scassellati, **Jake Brawer**, Katherine Tsui, Setareh Nasihati Gilani, Melissa Malzkuhn, Barbara Manini, Adam Stone, Geo Kartheiser, Arcangelo Merla, Ari Shapiro, et al. (2018). "**Teaching Language to Deaf Infants with a Robot and a Virtual Human**". In: *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, p. 553.

Zong Xuan Tan, **Jake Brawer**, and Brian Scassellati (2018). "**That's Mine! Learning Ownership Relations and Norms for Robots** ". In: *Thirty-second AAAI conference on artificial intelligence*.

Workshop proceedings

Ava Abderezaei, Chi-Hui Lin, Joseph Miceli, Naren Sivagnanadasan, Stéphane Aroca-Ouellette, **Jake Brawer**, and Alessandro Roncone (2025). "**Towards Zero-Shot Coordination between Teams of Agents: The N-XPlay Framework**". In: *Workshop on Multi-Robot Systems*. Robotics: Science and Systems (RSS). Workshop paper; proceedings forthcoming.

Jake Brawer, Kayleigh Bishop, Bradley Hayes, and Alessandro Roncone (2023). "**Towards A Natural Language Interface for Flexible Multi-Agent Task Assignment**". In: *Proceedings of the AAAI Symposium Series*. Vol. 2. 1, pp. 167–171.

AWARDS AND FELLOWSHIPS

July 2023, 2024, 2025 — **Army Educational Outreach Program (AEOP) Fellowship**, University of Colorado Boulder — awarded in 2023, 2024, and 2025 to support postdoctoral research.

March 2023 — **Best paper (technical track):** Our paper "Interactive Policy Shaping for Human-Robot Collaboration" won the best paper award in the technical track at the 2023 International Conference on Human-Robot Interaction (HRI 2023).

WORKSHOPS ORGANIZED

March 2023 — **Human-Machine Teaming Paradigm Workshop**
Jake Brawer, Bradley Hayes, Alessandro Roncone

March 2018 — **Bridging the Gap: An NSF Workshop on Conversational Agents and Human-Robot Interaction**
Justine Cassell, Brian Scassellati, Jake Brawer, Michael Madaio

NSF Cyber–Human Systems (CHS), Robust Intelligence, National Robotics Initiative. Award #1829237

ADDITIONAL TEACHING EXPERIENCE

Spring 2018/2019 — **Intelligent Robotics**
Teaching Assistant

Fall 2017 — **Object Oriented Programming**
Teaching Assistant

Fall 2014/2015 — **Perception and Action**
Teaching Assistant

SERVICE

Leadership — Area Chair, International Conference on Human–Agent Interaction (**HAI**; 2024)

Conference Reviews — **RSS** (2025 — **Outstanding Reviewer Award**), **HRI** (2017–2025 — **Outstanding Review Mention 2023**), **IROS** (2020), **ICRA** (2019), **Humanoids** (2018)

Journal Reviews — **IJRR** (2024), **T-RO** (2024), **THRI** (2019–2025)

Student Supervision — Shivendra Agrawal (2024–2025) Ava Abderezaei (2024–2025) Gyaning Kumar (2025) Yi-Shiuan Huang (2025) Chi-Hui Lin (2024–2025) Aaquib Tabrez (2023–2024) Kaleb Bishop (2017–2019, 2023) John Dallard (2021–2022) Sarah Widder (2017–2019) Kevin Choi (2018) Acshi Haggenmiller (2016–2017) Tan Zong Xuan (2017–2018)